Entity Relationship Diagram

KodeHub | Classroom operations, individual learning and family access | Version 1.0

Purpose

This document defines KodeHub's logical data model and the ownership boundaries needed to connect it to Kodecoon's consolidated data. It is prepared for coordination between the KodeHub builder and the analytics builder.

The core relationship

A center contains class-code groups and their scheduled time slots. Students join these groups, while each student's module and lesson progress remains individual. Students in the same time slot can study entirely different modules. A schedule change must preserve each learner's history.

One record for shared information

Reuse the consolidated student, center, class-code and enrollment records. The shared registry supplies canonical IDs. KodeHub submits operational edits through the agreed integration interface; analytics reads authorized records. Two applications must not keep competing editable copies.

How to read the ERD

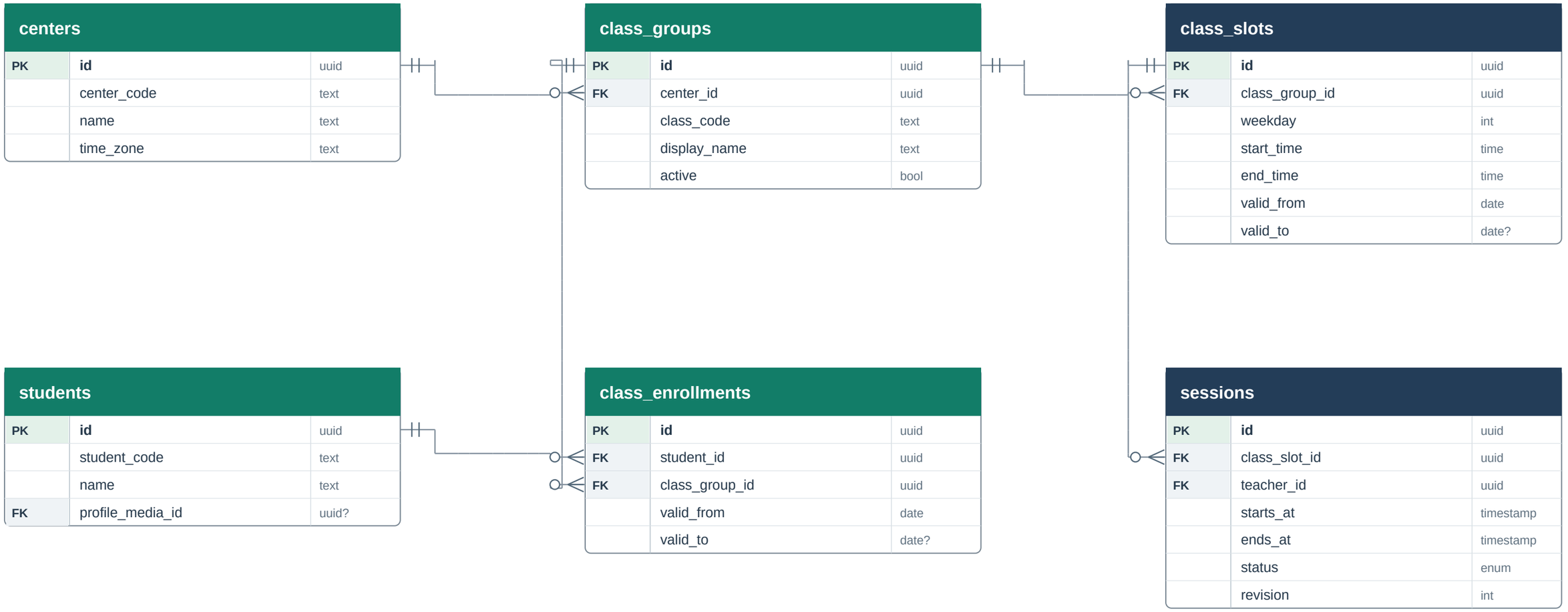
Each table lists its keys and principal fields. PK is its primary key; FK references another table. Crow's-foot connectors show relationship counts. Green tables belong to the shared registry; navy tables belong to KodeHub. Repeated tables marked [ref] are the same entity shown on another sheet. A question mark after a type means nullable.

Scope of the handoff

This is a logical interface specification. Physical table names and identifiers are mapped to the colleague's consolidated database during integration. It does not certify deployment. The companion relationship register identifies every FK, including links across diagram sheets. Financial amounts, receipts and bank details remain outside KodeHub.

01 / Centers, class codes and enrollment

Class-code identity and scheduling are separate from individual curriculum.



PK Primary key FK Foreign key || Exactly one o| Zero or one o< Zero or many

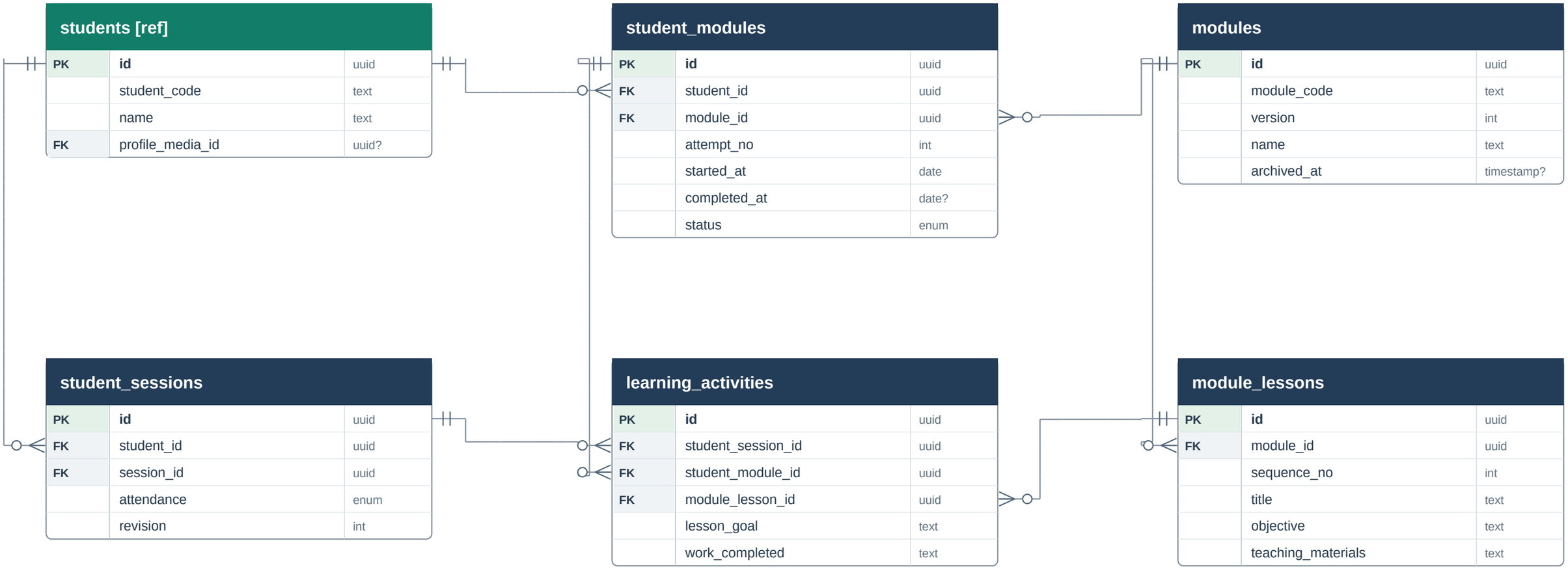
GREEN Shared registry NAVY KodeHub [ref] Entity repeated from another sheet

Keys: center_code is unique; class_code is unique within its center. An enrollment records a dated period in a class group. Slot validity records schedule changes; session start/end timestamps describe the actual occurrence.

Do not erase later enrollment changes when correcting an earlier date. Make-up attendance is recorded in student_sessions. The center time zone controls timetable display; the database stores actual session timestamps consistently.

02 / Individual modules and class activity

Different learners in one time slot can follow different modules and lessons.



PK Primary key FK Foreign key || Exactly one o| Zero or one o< Zero or many

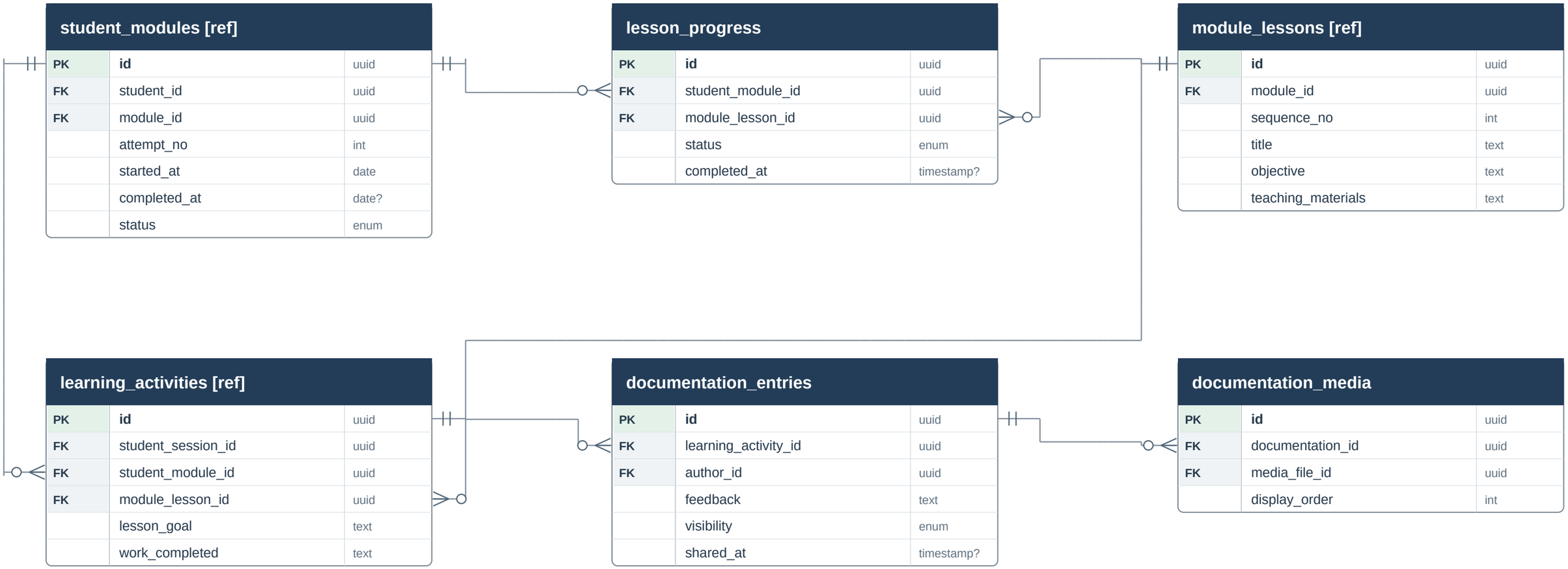
GREEN Shared registry NAVY KodeHub [ref] Entity repeated from another sheet

Keys: UNIQUE(student_id, module_id, attempt_no), UNIQUE(module_id, sequence_no), UNIQUE(student_id, session_id). A repeated module creates a new student_modules attempt. A session may contain several learning activities.

Validate that the activity's student_module belongs to the attending student, and its lesson belongs to that module. Moving the student to another class group must preserve module attempts, activities and learning history.

03 / Progress and class documentation

Learning records stay attached to the student, module attempt, lesson and class date.



PK Primary key FK Foreign key || Exactly one o| Zero or one o< Zero or many

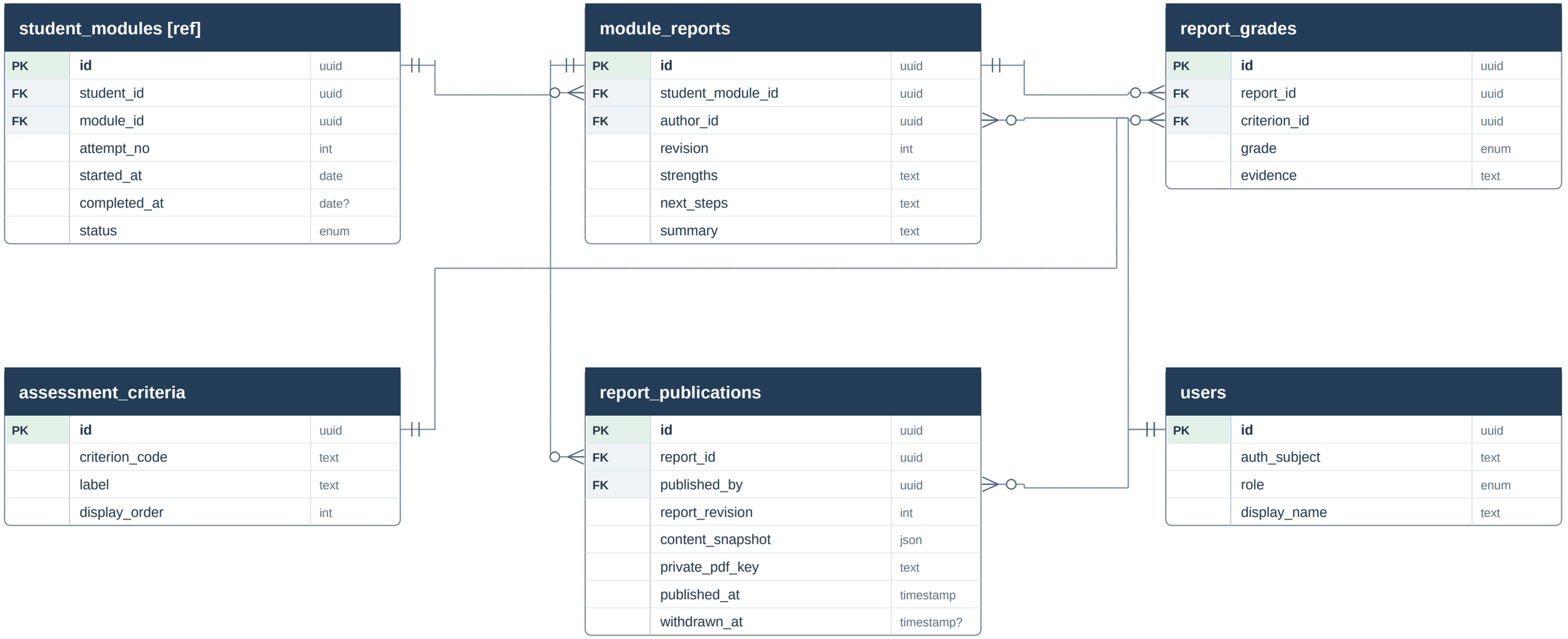
GREEN Shared registry NAVY KodeHub [ref] Entity repeated from another sheet

Keys: UNIQUE(student_module_id, module_lesson_id) for progress; UNIQUE(documentation_id, media_file_id) for attachments. Documentation contains short class feedback; media links attach Drive photos/videos without duplicating the underlying file.

Only shared documentation is available to linked families. Staff notes remain private. Module-end assessment is stored separately from class documentation. Media access must be checked against the documentation's learner.

04 / End-of-module assessment

Dedicated report generator: select learner/module, enter grades, preview, publish and export.



PK Primary key FK Foreign key || Exactly one o| Zero or one o< Zero or many

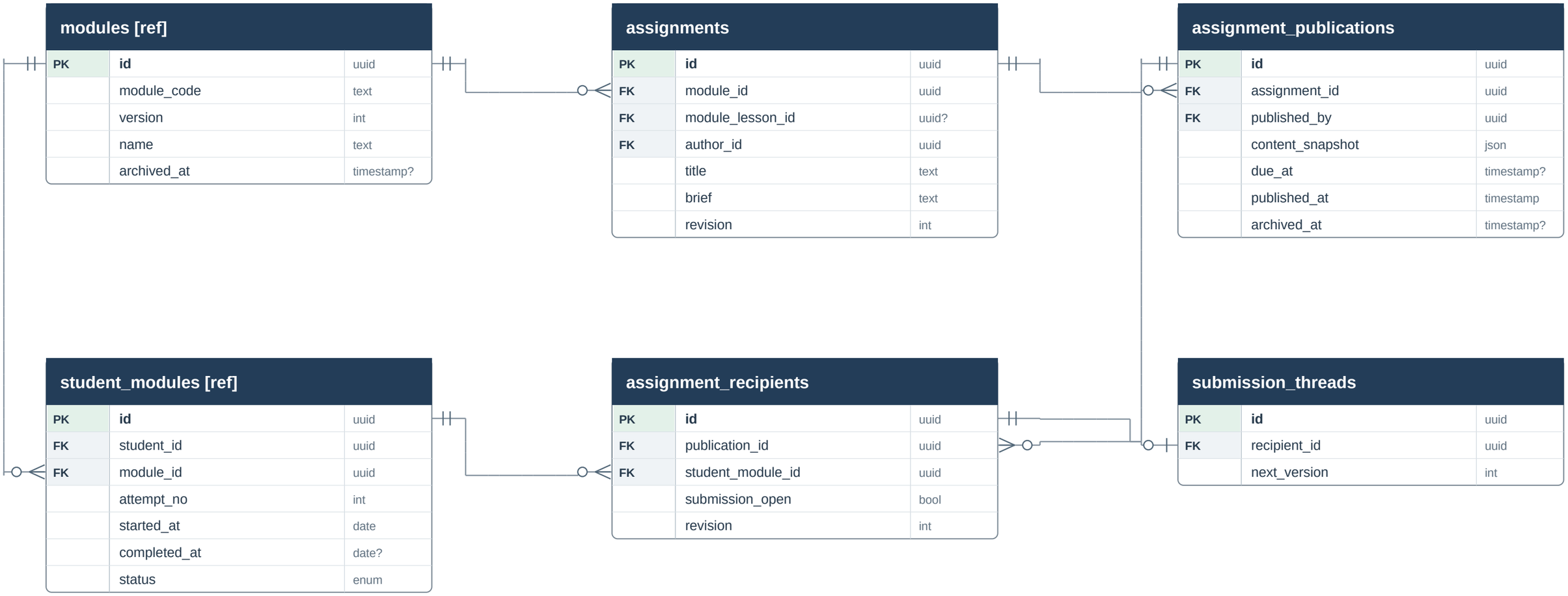
GREEN Shared registry NAVY KodeHub [ref] Entity repeated from another sheet

Keys: UNIQUE(report_id, criterion_id), UNIQUE(report_id, report_revision). Criteria: coding, building/designing, creativity, problem solving and communication. Grades: S, A+, A, A-, B+, B, B-. Unobserved criteria can be marked Not assessed.

Publication freezes names, module, criterion labels, grades, grade explanations and report text. The PDF includes the Kodecoon logo, student/module names and grade legend. Edits remain private until another revision is published; withdrawal preserves history.

05 / Assignments and learner recipients

Assignment membership follows an individual module attempt, with a captured published brief.



PK Primary key FK Foreign key || Exactly one o| Zero or one o< Zero or many

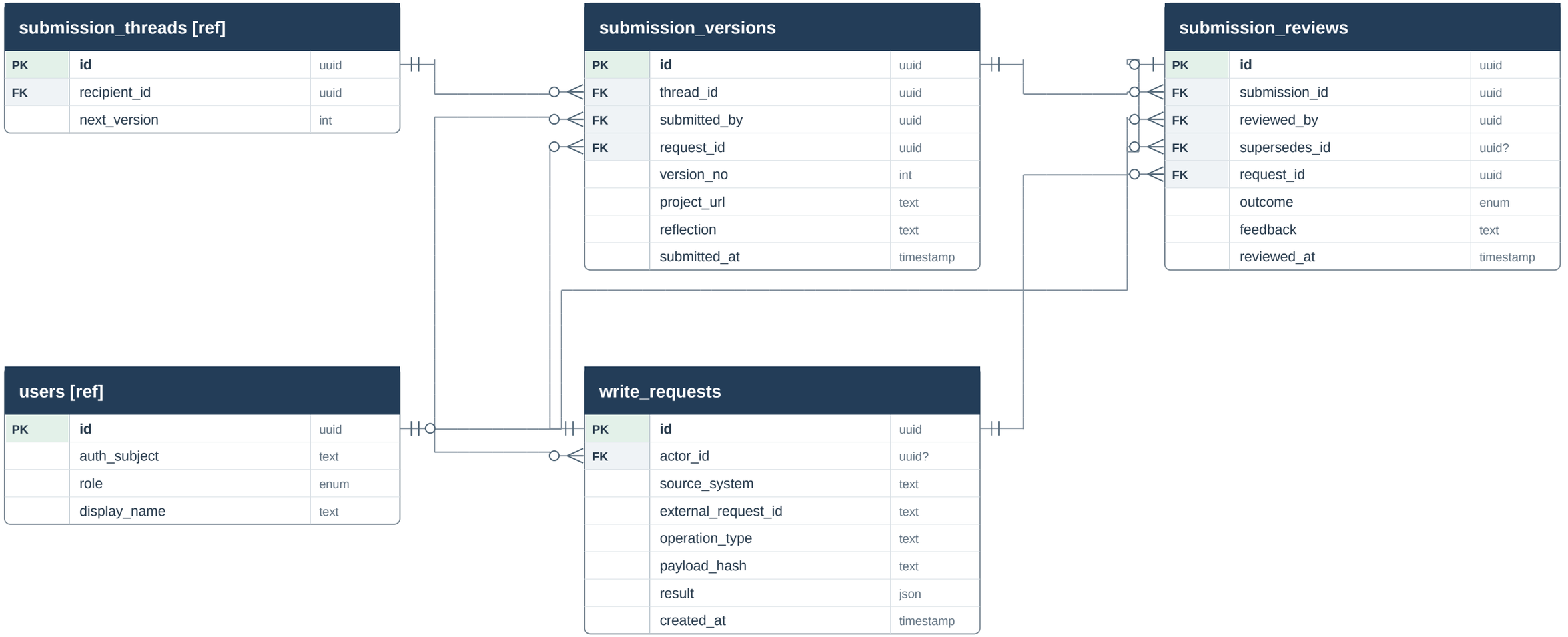
GREEN Shared registry NAVY KodeHub [ref] Entity repeated from another sheet

Keys: UNIQUE(publication_id, student_module_id); UNIQUE(recipient_id) for the submission thread. Publishing captures assignment content, due date and intended recipients. A lesson reference must belong to the assignment's module.

Recipients retain their history when class times change. Closing submission permission does not delete work. Reopening and adding recipients are explicit staff actions with revision checks. Families act only for a linked child.

06 / Submission and review history

Immutable work versions and teacher review corrections preserve the full assessment record.



PK Primary key FK Foreign key || Exactly one o| Zero or one o< Zero or many

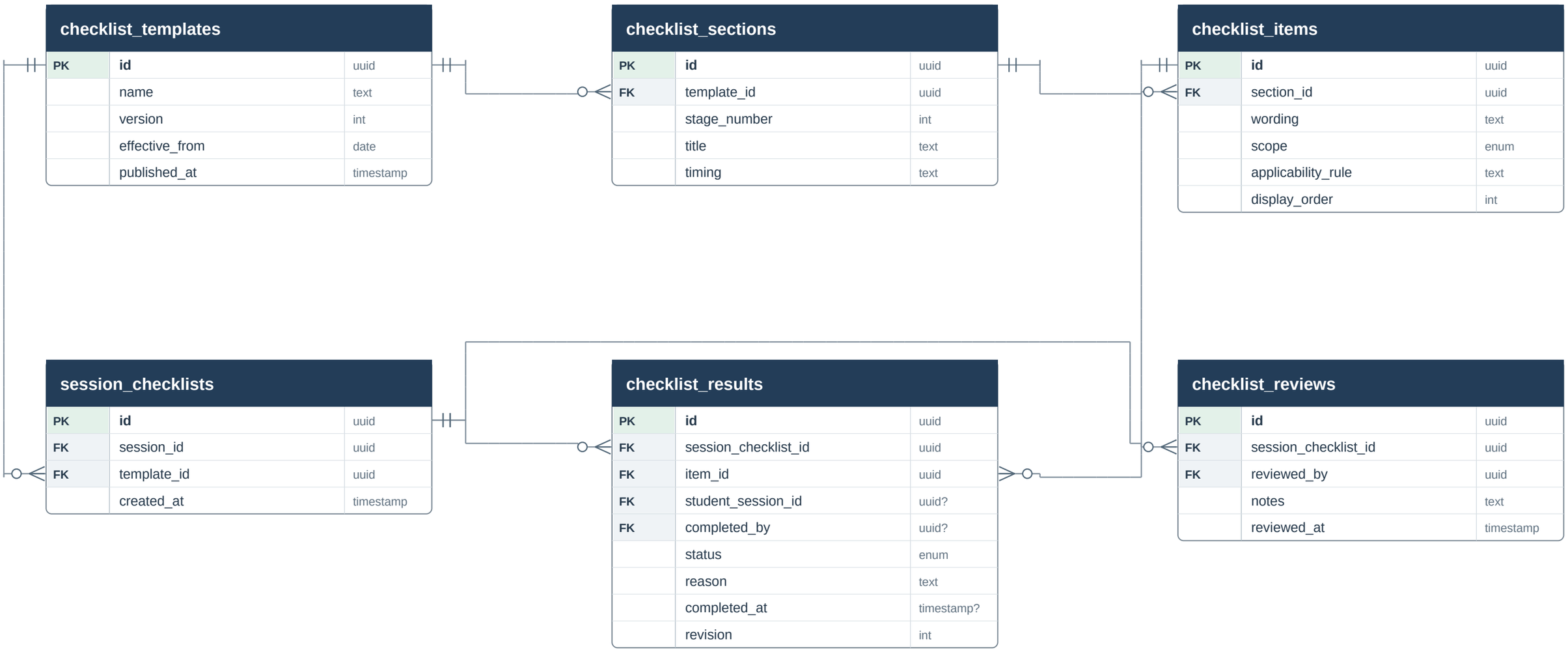
GREEN Shared registry NAVY KodeHub [ref] Entity repeated from another sheet

Keys: UNIQUE(thread_id, version_no) and UNIQUE(source_system, external_request_id). A correction appends a review linked to the previous review; supersedes_id cannot point outside the same submitted version.

Allocate versions and check the latest version transactionally. Requests carry a unique operation ID: the same payload retries safely, while reused IDs with different payloads conflict. An attributed guardian/staff actor submits the work; nobody impersonates a learner.

07 / Teacher checklist and supervision

Class tasks count once; individual-student follow-ups count per applicable learner.



PK Primary key FK Foreign key || Exactly one o| Zero or one o< Zero or many

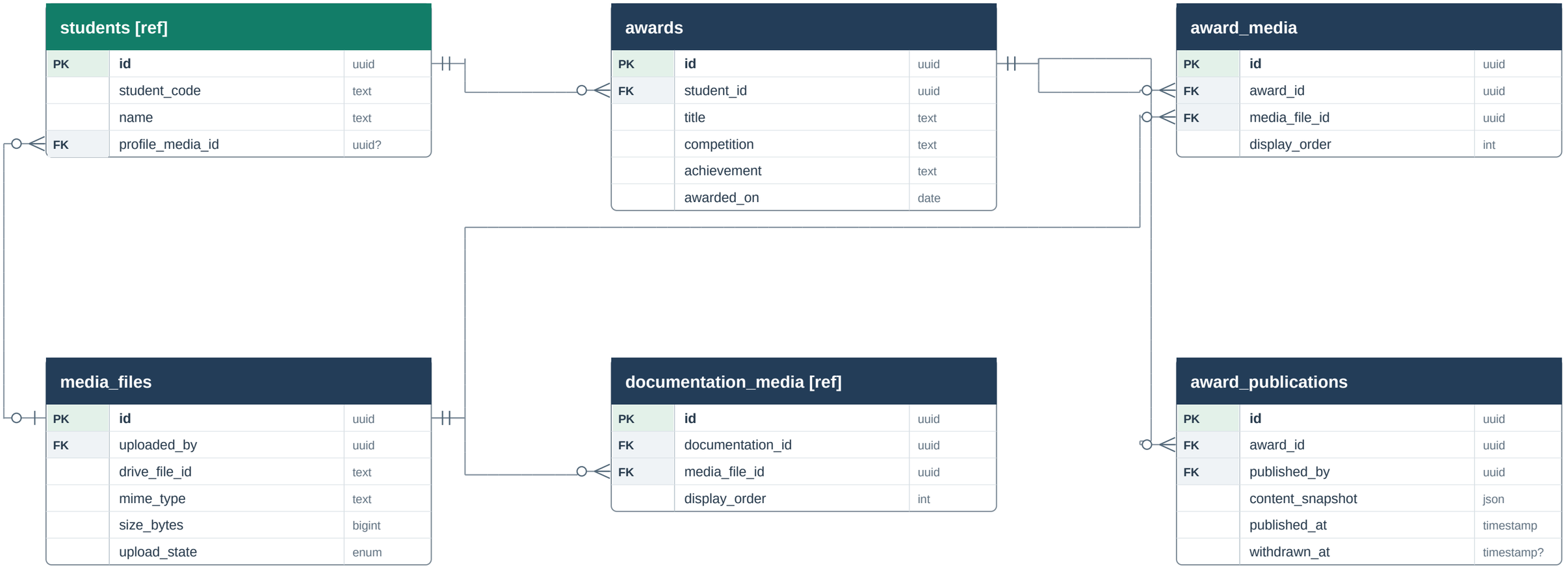
GREEN Shared registry NAVY KodeHub [ref] Entity repeated from another sheet

Source checklist: 38 checkboxes across eight stages (7/4/5/8/3/5/3/3). Pin a published template version per session. student_session_id is null for class tasks and required for student tasks; enforce one result per applicable item/learner.

Completion = done / applicable required items. N/A requires a reason and is excluded. Pending, overdue and not-yet-due remain distinguishable. Supervisor views compute center -> time slot -> student detail from these records, without a second editable checklist copy.

08 / Photos, videos and Hall of Fame

Academy-controlled Google Drive stores student media; KodeHub stores access-scoped file references.



PK Primary key FK Foreign key || Exactly one o| Zero or one o< Zero or many

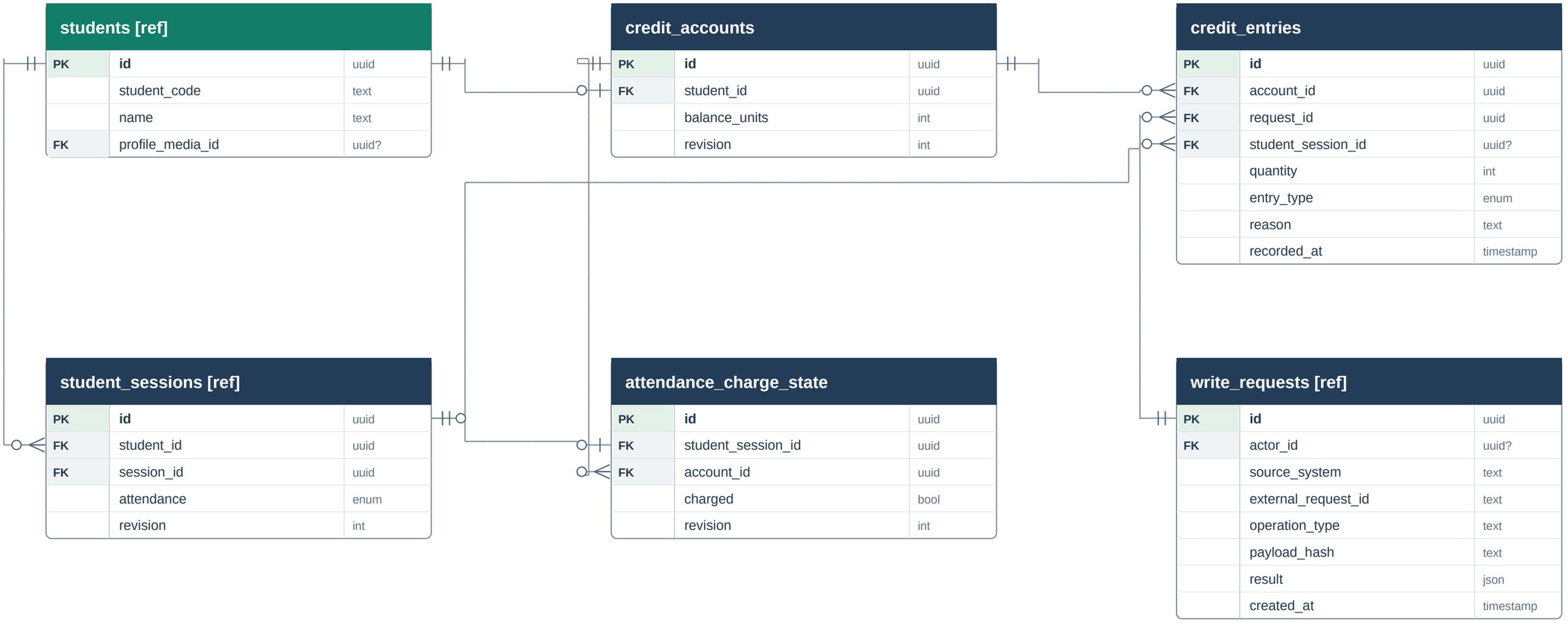
GREEN Shared registry NAVY KodeHub [ref] Entity repeated from another sheet

Keys: UNIQUE(drive_file_id), UNIQUE(award_id, media_file_id). A single stored file can be linked to documentation or an award. students.profile_media_id identifies the profile portrait. An upload must be finalized before family access.

Private playback must work for a parent logged into KodeHub without the teacher's Google login. Authorize every file from its linked child and shared record. Award publications expose selected achievement content and associated album images; staff retain editing controls.

09 / Credit units and attendance charges

One authoritative credit ledger; payment amounts and banking information stay external.



PK Primary key FK Foreign key || Exactly one o| Zero or one o< Zero or many

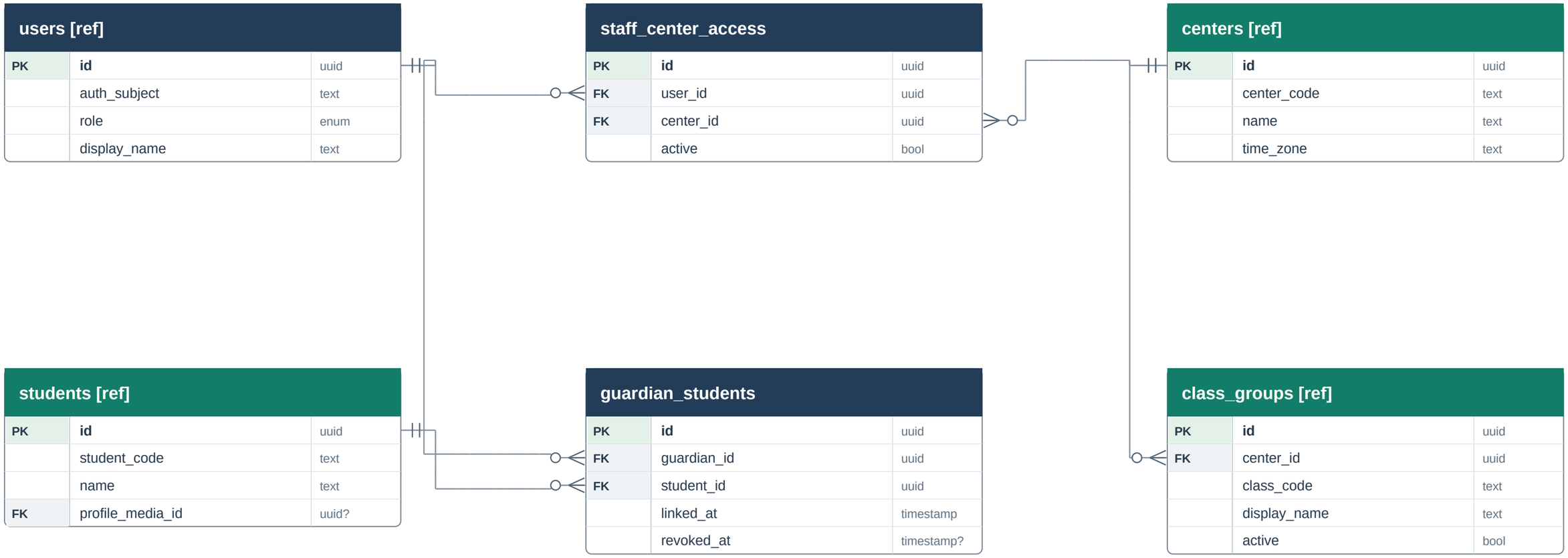
GREEN Shared registry NAVY KodeHub [ref] Entity repeated from another sheet

Keys: UNIQUE(student_id) for accounts and UNIQUE(student_session_id) for charge state. balance_units must equal the ledger sum and stay non-negative. Ledger entries are append-only; corrections add compensating entries.

An authorized finance grant sends canonical student ID, credit units and a unique operation reference, without payment details. KodeHub records attendance use/refunds atomically with the balance and request receipt. A retry must never create a second deduction or grant.

10 / Staff and family authorization

Teacher and center manager have equal staff permissions; family access is child-specific.



PK Primary key FK Foreign key || Exactly one o| Zero or one o< Zero or many

GREEN Shared registry NAVY KodeHub [ref] Entity repeated from another sheet

Keys: UNIQUE(auth_subject), UNIQUE(user_id, center_id) and at most one active guardian/student link. Staff-center membership grants the center access needed for the academy-wide timetable. Both staff roles use the same permission set.

Parents can view their linked child's shared documentation, progress, awards, credit-unit history and published reports, and submit work where allowed. Deny peer student access, private notes, staff checklists and finance data in pages, APIs and database rules.

11 / Responsibility for data changes

One authoritative record per topic; separate apps can use the same data safely.

Data	Authoritative home and write responsibility	Analytics / family access
Students, centers and class codes	Shared registry using the colleague's consolidated canonical IDs. KodeHub submits validated additions/edits through the integration interface. Shared core retains identity ownership.	Analytics reads the shared records. Families receive permitted fields for linked children only.
Enrollment and schedules	Shared enrollment history; KodeHub manages staff scheduling actions and dated sessions. Reconcile class codes and schedule records once; avoid duplicate editable rosters.	Analytics reads dated records and aggregates. Family profile shows the child's enrollment history.
Modules, learning and assessment	KodeHub owns module definitions, learner-module attempts, lesson progress, assignments, submitted work, reviews and module reports.	Analytics receives approved learning views. Families receive only their child's permitted/published records.
Media and awards	KodeHub owns media metadata and achievement records; the academy-controlled Drive owns the files. Access follows the linked child and shared content.	No direct unrestricted Drive folder listing. Analytics receives only metadata it needs.
Teacher checklist	KodeHub owns template versions, completion results and supervisor reviews. The dashboard calculates completion from those results.	Staff-only. Analytics can read approved completion aggregates.
Credit units	One ledger managed by the KodeHub credit writer. Finance sends authorized grant commands; attendance and staff corrections use controlled ledger operations.	Families see credit units and usage. No amount paid, receipts or bank details enter KodeHub.
Financial history	Finance/analytics domain outside KodeHub's database access. No KodeHub payment processing or finance tables.	Only the separately authorized financial system and analytics account may read financial records.

These boundaries define what KodeHub needs and writes. Map shared-core entities to the colleague's actual tables rather than creating copies. Each application uses its own least-privilege account; do not share a broad service-role credential.

12 / Integration contract

Information needed to connect the two applications without rebuilding the old duplication.

Canonical identifiers

Exchange the consolidated schema, primary keys, class-code rules and legacy-ID mappings. Reuse canonical student/center/class IDs; never match students by name alone. KodeHub-specific records reference those IDs. Guardian authorization comes from verified account links, never email similarity.

Write ownership and change delivery

Agree the endpoint or controlled database operation for each shared-core edit. Assign operation IDs, revision checks and an audit actor to changes. Analytics reads approved views or replicas; it does not independently rewrite the classroom records. Derived dashboards never become a second source of attendance or balances.

Credit opening position and cutover

Reconcile opening credit units and active enrollment dates before daily use. Deduplicate imported operations by source identifier. Once a record moves to the shared system, stop editing the corresponding Tracker/OClass copy or explicitly designate it as an output-only copy.

Relationship register

The accompanying relationship-register.csv lists every FK, its target and whether it can be absent. Entity dictionaries list the fields shown in the diagrams. Domain sheets repeat reference tables so each diagram remains readable; repeated names always identify the same table.

Implementation boundary

This document is the KodeHub logical data contract for integration. It makes no claim that database migrations or live authorization have already been deployed. Physical integration mapping follows comparison with the consolidated schema; no student data or credentials are needed to exchange the ERD.